Traffic Speed Prediction

Mini-Project Overview — A quick guide for a new project member

1. What is the project about?

The project focuses on **Short-Term Traffic Speed Prediction**. The objective is to use historical traffic data to predict the speed of vehicles on roads in the near future — for example, the next 10, 20 or 30 minutes.

Simple example:

8:00 → 45 km/h 8:10 → 42 km/h 8:20 → 38 km/h 8:30 → 35 km/h
The model learns such patterns and predicts the likely speed at 8:40.

2. What data do we have?

We are working with the **EXPY-TKY Traffic dataset**, containing traffic-speed observations for many road links in Tokyo. The main traffic data contains a timestamp, road/link ID and recorded speed, with observations at approximately 10-minute intervals.

The project also has road-relationship information (adjacency/distance matrices), which can help represent how different road links are connected or related.

3. What are we going to build?

Model	Purpose
XGBoost	Traditional ML baseline for comparison
LSTM	Learns how traffic changes over time
STGCN	Learns both road relationships and time patterns
iTransformer (optional)	A modern time-series model to investigate if feasible

4. Why use different models?

Traffic is both a **time-series** problem and a **spatial** problem. The speed of a road depends not only on its previous speeds but can also be affected by nearby or connected roads.

So we will compare a traditional ML approach (XGBoost), a temporal deep-learning approach (LSTM), and a spatio-temporal approach (STGCN). This comparison gives the project a stronger research component than using only one basic model.

5. Planned workflow

Raw data → Cleaning & preprocessing → Time-series sequences → Model training → Prediction → Evaluation → Model comparison → Analysis

6. What will we compare?

We will evaluate the models using metrics such as **MAE** and **RMSE**, and compare performance across different prediction horizons (for example, 10/20/30 minutes). We will also study whether including road-to-road relationships improves prediction.

7. Work that needs to be done

Stage	Main work
Data	Understand, clean and prepare the traffic data
Features	Create historical time windows and prediction targets
Baseline	Train and evaluate XGBoost
Deep Learning	Implement/train LSTM and STGCN
Experiments	Test different horizons and settings
Analysis	Compare results and understand model behaviour
Report/Paper	Graphs, tables, methodology, results and documentation

8. In one sentence

We are building a system that learns from past traffic conditions and road relationships to predict near-future traffic speed, then comparing traditional ML, temporal deep learning and spatio-temporal deep learning approaches.

You do not need to know all the models before joining. The work can be divided between data preprocessing, model implementation, experiments and analysis.